# Transfer: momentum and impulse

**antocciphysics**



**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Apply momentum reasoning to new collisions and deflection cases.

**SUCCESS CRITERIA - I can...**

1. Define the system and positive direction.
2. Use conservation or impulse only when assumptions apply.
3. Explain energy and model limits clearly.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

**LANGUAGE GOAL** Use “momentum” and “impulse” to explain your model or result. Sentence starters are optional.

**momentum**: mass times velocity, including direction

**impulse**: net force acting over time; change in momentum

**conserved**: unchanged in total under the stated conditions

## 01 Bring back an earlier idea

A toy target needs a 3 km sideways shift over $5\times10^{6}$ s. Its mass is $7\times10^{8}$ kg.
**Estimate** Δv and impulse; state the model used.

**Word help:** estimate = find an approximate value

**Sentence starter:** I found ... [unit]. This value means ... in this situation.

**Marzano check - circle:** ① ② ③

## 02 Explain a worked example

**Reference:** $p = mv$; $J = \Delta p = F_{avg}\,\Delta t$; $p_{total,\,i} = p_{total,\,f}$.
Use momentum conservation only when external impulse is negligible. Label before/after diagrams and directions.

**Worked reasoning:** A 2 kg cart moving at 3 m/s stops in 0.5 s. Momentum change is -6 kg·m/s; average force is -12 N. Force direction opposes the initial motion.

**Your turn:** Represent the same stop with momentum arrows and a force-time rectangle.

**Sentence starter:** The ... in my model represents ... . Its label shows ... .

**Marzano check - circle:** ① ② ③

## 03 Represent and interpret

A 1 kg cart at 4 m/s right sticks to a 1 kg cart at rest.
**Draw** before/after and explain which quantity is conserved.

**Word help:** at rest = not moving in the chosen reference frame

**Sentence starter:** Under the assumption ..., the evidence ... supports ... because ... .

Before contact: add velocity arrows.




**Marzano check - circle:** ① ② ③

## 04 Solve and justify

A 2 kg cart at +6 m/s sticks to a 1 kg cart at -3 m/s.
**Find** final velocity and kinetic energy transformed.

**Sentence starter:** I found ... [unit]. This value means ... in this situation.

**Marzano check - circle:** ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

## 05 Analyze the reasoning

A report uses equal total momentum before/after to claim a collision is elastic. What additional calculation is required?

**Sentence starter:** I notice … . This supports … because … .

**Marzano check - circle:** 1 2 3

## 06 Choose a method

A cart has momentum -24 kg·m/s and mass 4 kg.
**Find** velocity and its new momentum if velocity reverses at unchanged speed.

**Word help:** velocity = rate of position change, with direction

**Sentence starter:** I found … [unit]. This value means … in this situation.

**Marzano check - circle:** 1 2 3

## 07 Exit check

A triangular force pulse has peak 40 N and duration .30 s. It acts on a 2 kg cart initially at -1 m/s.
**Find** impulse and final velocity.

**Sentence starter:** I found … [unit]. This value means … in this situation.

**Marzano check - circle:** 1 2 3

## 08 STRETCH: Apply it another way

A 0.4 kg ball rebounds from +5 to -3 m/s. A triangular force pulse lasts 0.08 s.
**Find** impulse, average force and peak-force magnitude.

**Word help:** magnitude = size of a quantity without its direction

**Sentence starter:** I found … [unit]. This value means … in this situation.

**Marzano check - circle:** 1 2 3